Hans Wassermann

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Education

University of Michigan Aug 2026 – May 2028
Master of Science in Electrical and Computer Engineering *Ann Arbor, MI*

- Control Systems Specialization

Rutgers University Sep 2023 – May 2026
Bachelor of Science in Electrical and Computer Engineering | 3 Year Track *New Brunswick, NJ*

- **GPA:** 4.00 / 4.00, Dean's List (All Semesters), Engineering Honors Academy, Learning Assistant, Matthew Leydt Society
- **Relevant Coursework:** Control System Design, Linear Systems and Signals, Digital Signal Processing, Embedded Systems

Experience

Controls Graduate Research Assistant Sep 2026 – Present
Precision Systems Design Lab | MATLAB, Simulink, dSpace *Ann Arbor, MI*

- Conducting **system identification** via experimental frequency testing on a 2-axis flexure-based nanopositioning stage
- Designing **PID and feedforward controllers** in MATLAB/Simulink to achieve fast **point-to-point positioning**

Undergraduate Research Assistant — Robotics & Controls May 2026 – Aug 2026
SWARM Intelligence Lab | MATLAB, Python, C, Git, OpenCV, Serial Communication *New Brunswick, NJ*

- Developed a **multi-agent motion planning system** in Python to steer 4 magnetic fish across a 75x75 solenoid grid in real time, using camera feedback at 20Hz and a lookahead buffer to follow paths from a **physics-based schooling simulation**
- Implemented **inter-agent repulsion dynamics** in the schooling model to enforce a hard minimum spacing constraint across all agents, preventing collisions while preserving emergent collective behavior
- Designed stall detection and recovery logic using **camera-based position feedback**, including waypoint-skip and coil-reanchor strategies validated through log analysis

Augmented Reality Research Intern May 2025 – Aug 2025
Rutgers WINLAB | C#, Unity, OpenCV, Python, MQTT, GitLab, Linux *New Brunswick, NJ*

- Extended a **multi-user collaborative AR** drawing platform by building avatar and username systems, syncing user positions and movement across HoloLens clients in real time using **MQTT**
- Designed a **QR-code-based spatial anchoring system**, integrated into the app's launch procedure, to **synchronize coordinate frames** across multiple AR headsets
- Prototyped a **real-time body-tracking pipeline** using an Intel RealSense depth camera and **YOLO pose estimation**, serializing pose data over MQTT to reconstruct user movement on other clients

Data Engineering Intern May 2024 – Aug 2024
BASF ECMS | Python, Jupyter Notebook, Excel, VBA *Iselin, NJ*

- Automated **real-time data processing for 20+ reactor test protocols in Python**, cutting manual analysis time
- Built **automated quality-control checks** into the pipeline, flagging out-of-range sensor readings before reaching engineers
- Developed an **interactive dashboard** for engineers to access and visualize raw and processed test data

Projects & Clubs

Michigan Autonomous Aerial Vehicles Club – GNC Team | *ROS 2, Gazebo, SITL, C, Python, Linux* Sep 2026 – Present

- Set up a **PX4 SITL + Gazebo quadrotor simulation**, integrating flight telemetry with **ROS 2** via Micro XRCE-DDS
- Refactored **waypoint-following logic** to support heading-based arrival criteria and timed hold states before advancing

Beehive Monitoring System | *C++, Python, PlatformIO, FastAPI, PyTorch, vLLM, Onshape* Sep 2025 – May 2026

- Designed a dual-ESP32 **embedded sensing system** for autonomous beehive field deployment
- Interfaced and calibrated a **MEMS microphone**, BME280 sensor, and **4 load cells**, with protoboard assembly and a **3D-printed enclosure** for field hardware
- Built the **BLE/WiFi telemetry pipeline** connecting embedded hardware to a full-stack backend, and **integrated teammates' ML audio classifier** and LLM chatbot into the live sensor pipeline for natural-language hive querying

Neuron Model Control System Design | *MATLAB, Simulink, Nonlinear Control, Automated Testing* Sep 2025 – Dec 2025

- Linearized the nonlinear Hindmarsh-Rose neuron model about analytically computed equilibria, performed **Jacobian-based stability analysis** to classify fixed points, and validated dynamics against theoretical outputs in MATLAB/Simulink
- Designed and tuned **state-feedback and PID controllers** to regulate neuronal firing rate, using root locus and frequency-domain methods to meet **closed-loop performance** specifications

Technical Skills

Programming: Python, MATLAB, C, C++, C#, Java
Embedded Systems: SPI, I2C, I2S, BLE, Zynq SoC/FPGA, Vivado, Vitis, ARM Cortex-A9, Embedded Linux, PlatformIO
Hardware & Test: Oscilloscope, Function Generator, Multimeter, Power Supply, Soldering, LTspice, KiCAD
Tools: ROS 2, Git, Simulink, Linux, Onshape, Jupyter Notebook, Gazebo, Unity, dSpace, Microsoft Office